

When does self-duality cause physics?

A machine-checked criterion, tested across fifteen candidate physical systems, an interventional experiment, and real cosmological data

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Abstract

Physics is full of self-dual points: places where a symmetry exchanges two descriptions of the same system and maps it to itself. Some of these points are physically decisive – the Kramers–Wannier point fixes the Ising critical temperature exactly – and some are not: the free boson’s self-dual radius is an ordinary point on a line of conformal field theories, not a phase boundary. This paper states, proves in Lean 4 against Mathlib with two independent kernels, and tests experimentally, a single criterion that predicts which is which: a self-dual point is forced to be physically distinguished if and only if it exchanges two *inequivalent* sectors on a *discrete* fixed-point set; on a continuous moduli space with enhanced symmetry, or when self-duality only constrains a fixed point without producing one, it is not. The duality is always exact; what does causal work, when anything does, is the *sector* it relabels – a conserved or topological label crossing a threshold, proliferating, annihilating, or being imposed. We test this criterion, not just state it, in three independent ways. First, a machine-checked library of 257 theorems across 30 Lean modules states the mathematical skeleton of fifteen candidate cross-domain cases – from the Ising model to Calabi–Yau mirror symmetry – and every module is re-checked by a second, independent kernel (nanoda) so that a statement’s acceptance does not rest on one implementation of type theory. Second, a pre-registered, energy-matched intervention in a real classical field (a projected Gross–Pitaevskii superfluid) asks whether a sector is a *cause*, not merely a correlate: on three equilibrated bases, injecting the same energy as topological defects lowers the condensate fraction by 0.49–0.73 where injecting it as phonons lowers it by at most 0.035 – the same energy, as topology, is 15–60× more effective. Third, the criterion is carried into cosmology (dark matter halos, dark energy, the K3 charge lattice of a black hole’s dyonic charges) and against one real, present-day dataset: an independent, exact reproduction of a published CMB bound on a discrete dark-energy phase transition, built from the published equations and Planck’s own data, agrees with the original to within a factor of 3.16 across the parameter grid we tested, and a forecast built on that same reproduction shows that no future CMB temperature experiment can meaningfully tighten this particular bound, since Planck’s own measurement is already within about 15% of the cosmic-variance floor at the relevant multipoles. Eleven of these fifteen candidates receive one of the criterion’s four verdicts; three are, on inspection, not instances of the criterion at all – a classification (topological superconductors), a duality between different theories with no self-map (three-dimensional Ising duality, AdS/CFT) – and one, Calabi–Yau mirror symmetry, is left an open question rather than forced into a verdict the literature does not support. Reporting these honestly is part of the theory’s claim to be checked, not merely asserted.

1 The puzzle

A self-dual point is a place in a theory’s parameter space where an exact symmetry exchanges two descriptions of the same physics and maps the theory to itself. The symmetry is called a

duality: strong coupling for weak, high temperature for low, one radius for its inverse, one gauge group for another. Dualities of this kind recur across essentially every part of theoretical physics, and a self-dual point – the fixed locus of the exchange – is often treated, informally, as if it must be special: a natural place to look for a phase transition, a natural place to expect new physics.

It is not always special. The Kramers–Wannier point of the two-dimensional Ising model [1] is exactly the critical temperature: the order and disorder operators [2] are exchanged by the duality, and $\mathbb{Z}_2$ has exactly one symmetric point, so the two phases the duality exchanges are forced to meet there. But the free boson’s self-dual radius $R = \sqrt{2}$ under T-duality $R \leftrightarrow 2/R$ is not a phase boundary at all – it sits on a continuous line of conformal fixed points, one for every R , and the point where a vortex operator actually becomes relevant (the Berezinskii–Kosterlitz–Thouless transition [3]) is a different radius entirely, forced by a threshold condition on one sector, not by the duality’s fixed point. The same pattern – sometimes decisive, sometimes not – recurs in string-gas cosmology, the superconductor–insulator transition, and the quantum Hall effect, and, as this paper documents, in supersymmetric gauge theory, $\mathcal{N} = 4$ super Yang–Mills, rational conformal field theory, the fractional quantum Hall effect, and (with an answer this paper declines to give, on purpose) Calabi–Yau mirror symmetry.

The question this paper answers is not “is this duality real” – in every case examined, it is, exactly, and none of the underlying mathematics is new. The question is: *when is a self-dual point a cause of anything, and when is it merely a coincidence of bookkeeping?* We answer it with a single criterion, state the criterion’s content as machine-checked mathematics wherever the checkable content exists, and test the criterion’s most basic presupposition – that the sector, not the duality, is where the causal weight lies – with a real experiment.

The criterion. A self-dual point pins a physical transition if and only if the two sectors it exchanges are simultaneously critical and the fixed-point set is discrete. On a continuous moduli space with enhanced symmetry, the self-dual point is a point like any other, and any transition present sits where one sector crosses its own threshold – not at the duality’s fixed locus. When both dual sectors are populated and compete energetically, the self-dual point can be a preferred equilibrium rather than a transition. In every case, the duality relabels the sectors; the sector acts.

2 Sector, duality, self-dual point: the formal skeleton

Every case in this paper has the same shape. There is a *sector*: a discrete or continuous label – a winding number, a Landau-level filling, a coupling constant, a gauge group rank, a point in a moduli space – that indexes physically distinguishable configurations or theories. There is a *duality*: an involution or a Fourier-type reindexing map on the sector, that is exact, invertible (or, in the non-invertible cases the modern literature has clarified, at least an honest map between two descriptions), and preserves the physical content – the spectrum, the partition function, the correlators – while relabelling which sector produces it. And there is, sometimes, a *self-dual point* or *locus*: a fixed point of the duality map.

The mathematics of the reindexing step is, in every clean case, literal Fourier analysis on a group of sectors [4]: the compact boson’s $(n, w) \mapsto (w, n)$ under $R \mapsto 2/R$; the Ising model’s Kramers–Wannier map, itself a Fourier transform on $\mathbb{Z}_2$; the discrete Fourier transform on $\mathbb{Z}_N$ underlying a cyclic gauging; the modular S -matrix of a rational conformal field theory, a Fourier transform on the fusion ring exactly when that ring is abelian. Where this project’s own Lean library builds directly on Mathlib’s finite-group Fourier machinery (`ZMod.dft`, `ZMod.dft_dft`) rather than re-deriving discrete Fourier inversion, and where Mathlib’s own number-theory library already proves the modular-group stabilizer facts a Montonen–Olive S -duality case needs, the point is the same each time: the reindexing is elementary, checkable mathematics, and its elementariness is exactly what makes the distinction from physical causation possible to state

precisely.

What decides the verdict

Four structurally distinct outcomes recur across the eleven cases examined in this paper, and distinguishing them precisely is the paper’s central technical content:

1. **Forced.** The self-dual point sits on a *discrete* fixed-point set and exchanges two *physically inequivalent, gapped* sectors. The two phases the duality connects cannot both be continuously deformed into each other away from that point, so they are forced to meet there. Ising is the paradigm case; $SO(3)$ Yang–Mills at $\theta = \pi$, where the self-dual point separates a $\mathbb{Z}_2$ topological phase from the trivial phase, is a second, established in the modern non-invertible-symmetry literature [5].
2. **Not forced.** The self-dual point sits on a *continuous* moduli space of equivalent theories, at a point of *enhanced* (but not qualitatively different) symmetry. Nothing is caused there; the enhancement is a fact about the point’s stabilizer, not a transition. This is the modal outcome in this paper: the compact boson’s self-dual radius, the free-boson-to- $\widehat{su}(2)_1$ point of rational conformal field theory, the point $N_c = N_f/2$ of Seiberg duality where the electric and magnetic gauge groups coincide, and the point $\tau = i$ (and $\tau = \rho$) of Montonen–Olive S -duality in $\mathcal{N} = 4$ super Yang–Mills, all receive this verdict, each independently verified against its own primary literature.
3. **Constraint, not cause.** Self-duality, *if* the critical point happens to be self-dual, forces an exact numerical relation – a universal resistance, an exact transport coefficient – without itself causing the transition. The superconductor–insulator transition [6] and the particle-hole self-conjugate point $\nu = \frac{1}{2}$ of the fractional quantum Hall effect [7] both receive this verdict: the duality pins a number, the dynamics (vortex or charge proliferation; Fermi-liquid screening) causes the transition or the compressible phase.
4. **Organisation, not cause.** A duality can organise a diagram – the plateau structure of the quantum Hall effect under $\Gamma_0(2)$ [8], the class-number structure of black hole microstate counting under $SL(2, \mathbb{Z})$ on a $K3 \times T^2$ charge lattice [9] – while a completely different mechanism (Laughlin’s gauge argument [10]; the attractor mechanism fixing horizon moduli from charges) does the causing.

Three further outcomes are not verdicts of the criterion at all, and distinguishing them from a negative verdict is itself part of this paper’s discipline: a candidate can turn out, on reading the primary literature, not to be a duality (topological superconductors are a K-theory classification); a candidate can turn out to have no self-map to ask the question of (three-dimensional Ising duality relates a spin model to a *different* theory, a lattice gauge theory; AdS/CFT is a coupling-space equivalence with no conjectured involution); and a candidate can turn out to pose a question the literature genuinely does not answer (no citable self-mirror Calabi–Yau threefold exists to ask the question of, though the duality itself – mirror symmetry as T-duality on a torus fibration [11] – is real and verified). Reporting these three outcomes as what they are, rather than forcing them into the four-item taxonomy above, is not a limitation of the theory; it is a condition the theory imposes on itself.

3 Is the sector really a cause? An interventional test

Everything so far establishes a mathematical criterion for when self-duality is or is not *correlated* with a physical transition. It does not, by itself, establish that the sector is a *cause* in the sense that matters to an experimentalist: that intervening on the sector, holding everything else fixed,

changes the outcome. This is a distinct question, and this programme answers it directly, in the interventionist sense of Woodward and Pearl [12], with a pre-registered experiment in a real physical system rather than with a Lean theorem.

The system is a projected Gross–Pitaevskii classical field, a standard model of a two-dimensional Bose gas, prepared in three independent equilibrated base states. A machine-checked chain first establishes that the sector – the winding number of the phase around a loop – can carry an intervention at all: it is conserved by every continuous evolution that avoids a phase slip, a change of winding forces a slip, the slip on an XY ring costs a positive energy barrier for rings of at least ten sites (and the bound fails at exactly nine, the boundary case checked explicitly), the winding read at an enclosing scale R equals the net charge inside R so that a bound vortex–antivortex pair is invisible from outside, and the sampled, principal-branch loop sum used by every numerical vortex detector equals the topological degree of the continuum phase field for every sufficiently fine sampling.

The intervention itself is energy-matched: from each equilibrated base, four free vortex–antivortex pairs are injected at fixed particle number, momentum, and total energy, against a control arm that receives the identical energy as phonons instead. On all three bases, the condensate fraction falls by 0.49–0.73 in the vortex arm and by at most 0.035 in the phonon arm; the coherence exponent η grows by a factor of 9–44 in the vortex arm against at most 1.2 in the control. The same energy, delivered as topology rather than as thermal phonons, is fifteen to sixty times more effective at moving the same observables. The pre-registered composite claim is nevertheless *not* made in full: the vortex arm’s thermometer reads 5–16% *colder* than the phonon arm, the wrong sign to explain the effect by ordinary heating, and this is reported as a failure of the pre-registered criterion rather than reinterpreted after the fact. The honest reading, developed in a companion analysis and consistent with the point-vortex literature since Onsager [12], is that the injected vortex configuration is a subsystem with its own effective temperature, weakly coupled to the phonon bath – machine-checked at its combinatorial core (the inverse temperature of a state mixing two sectors is the *mediant* of the sector inverse temperatures, not their average, so equal energy does not imply equal temperature once a conserved label exists) and read together with two independent, calibrated thermometers on our own runs, giving the injected sector a temperature five to eight times the bath’s on three independent bases.

This is the paper’s load-bearing empirical claim: not that a duality relabels a sector (that part is elementary and machine-checked), but that the sector itself, independent of any duality relabelling it, is where an energy-matched intervention finds its effect.

What this experiment does not establish: that the thermometer discrepancy is fully explained (it is reported, not resolved); that this specific mechanism generalises beyond a two-dimensional classical field; or any claim about which of this paper’s cross-domain duality cases the intervention bears on directly – the interventional test establishes that *a* sector can be causally efficacious in *a* system, which is the presupposition the rest of this paper’s criterion needs, not a case-by-case proof that every sector discussed below is causally efficacious in its own system.

4 Fifteen candidates, eleven verdicts

Table 1 lists every cross-domain case examined under the criterion, with its verdict and the primary literature each verdict rests on. Full case-by-case argument, including the machine-checked Lean statement where one exists and an explicit account of what each Lean module does *not* prove, appears in the companion technical report [13] and its four dated addenda; this section summarises the verdicts and states what is genuinely new in the synthesis, namely the pattern across all eleven together.

Two features of this table are worth stating plainly, because they are easy to read past. First, of the fifteen candidates examined, only two – Ising and $SO(3)$ Yang–Mills – return the “forced”

verdict; the other nine cases that receive one of the criterion’s four verdicts return something else, and the remaining four rows fall outside the eleven-case verdict count for two different reasons: three because the criterion’s question does not apply to them at all (no duality, or no self-map), and one, Calabi–Yau mirror symmetry, because the literature does not resolve it. The modal outcome of applying this criterion honestly, across an intentionally wide sample of physics, is that a self-dual point is *not* a cause. Second, the criterion’s four positive categories were not fitted to the eleven verdicted cases after the fact: the compact boson, Ising, string-gas, and superconductor–insulator cases fixed the taxonomy in earlier work of this programme, and every case added since – quantum Hall organisation, RCFT, level-rank duality, Seiberg duality, Montonen–Olive S -duality, and the three exclusions – was classified by applying that fixed taxonomy to new primary literature, not by inventing a new category to fit. The one genuine addition to the taxonomy, forced by Calabi–Yau mirror symmetry, was not a new verdict but a new admission: that the literature can leave the question open, and that this is a legitimate outcome distinct from either a positive or a negative verdict.

5 Sectors in the sky: a cosmological extension, tested against real data

The criterion transfers, without new machinery, to three cosmological questions, developed at length in a companion paper [23] and summarised here for completeness.

Wave dark matter: the sector, not the duality, is what a density map can see. If dark matter is an ultralight scalar field, its condensate carries a topological winding sector exactly as a superfluid does. The question this programme’s own energy-matched intervention (§4) transplants to a cosmological halo is whether a density observable – the only one available to most surveys – can read a halo’s net circulation, or only its local vortex content. On our own torus, across twelve runs, a density-power observable is blind to an imposed sector at the 1–4% level while the same injected vortex pairs move it by 21–118%, an order of magnitude more – the torus transplant of an open question posed and left open by Zhou et al. (2025) and Brax & Valageas (2025) in real three-dimensional halos.

Dark energy: a walk on a sector lattice, and a test that separates a roll from a cascade. If the cosmological constant is a flux quantised on a lattice [24] or a discretely discharging scalar [25], its late-time evolution is a walk on an integer sector, not a continuous roll. The background expansion history alone cannot distinguish the two at current precision [26, 27]; a completed discrete transition, however, leaves a CMB-anisotropy signature with no analogue in a smooth roll, from the spatially random local completion time of a first-order phase transition [28, 29].

We did not stop at citing this test: we built it. Reading the published derivation directly – a bubble-nucleation power spectrum, a closed-form redshift-perturbation formula, and a Bessel-function projection onto the CMB temperature power spectrum, checked against Planck’s own published error bars by a three-bin χ^2 test, with no Boltzmann code and no Markov-chain Monte Carlo anywhere in the published pipeline – we independently re-implemented the entire calculation in plain numerical code against the real Planck 2018 temperature power spectrum, downloaded directly from the Planck Legacy Archive, first with an interpolated approximation to the underlying bubble-statistics spectrum and then, after locating the exact closed form in a companion paper’s supplementary material, with that exact spectrum in place of the approximation. The exact calculation closes most of the gap the first pass flagged as its weakest point – across the full grid of transition rates and completion redshifts we tested, the worst-case disagreement with the original authors’ own analytic approximation falls from a factor of sixteen

to a factor of 3.16 – but not uniformly: at low transition rates the exact result tracks the approximation closely, while at the highest rates we tested it now sits 30–45% *below* it, a real trade-off we report rather than reconcile, since the original approximation is itself described by its own authors as an order-of-magnitude estimate. We then asked whether any future CMB temperature experiment could improve on this bound at all, using the fact that this signal’s own statistic scales the bound exactly as the square root of the multipole error bar, with no linearisation needed: comparing Planck’s own measured errors against the pure cosmic-variance floor at the multipoles this signal occupies ($\ell \approx 5\text{--}50$) shows Planck is already within about 15% of that floor. No temperature-only successor to Planck – a bigger mirror, a colder detector – can meaningfully tighten this particular bound; what little room remains, at the very lowest multipoles, is capped by foreground and sky-masking systematics, not by instrument sensitivity, matching the original authors’ own pointer toward CMB spectral distortions and pulsar timing arrays as the more promising channels for further progress. Every step of this reproduction and forecast needed no specialised hardware: the full calculation runs in CPU-minutes to CPU-hours on a single core, confirming directly that a rented GPU would add nothing to this line of work, since every bound here is a closed-form or exact-scaling calculation, not a machine-learning or large-scale-simulation workload.

K3 as a charge lattice: the attractor mechanism, and where it does and does not generalise. A type II dyon on $K3 \times T^2$ has an entropy that is a function only of an $SL(2, \mathbb{Z})$ -invariant discriminant on its charge lattice [9]; the same discriminant, unmodified, governs the exact quarter-BPS degeneracy of a 4D-lifted D1-D5-P black hole once a fourth charge is added [16] – the $SL(2, \mathbb{Z})$ -invariance theorem transfers verbatim to a second physical setting. What does *not* transfer is the clean statement that entropy is a function of the sector alone: in the 4D-lifted setting an extra discrete invariant and wall-crossing are required [30], and the original Strominger–Vafa/Callan–Maldacena entropy formulas sit entirely outside this discriminant framework, being a different algebraic object (a product of three independent charges, not a quadratic-form discriminant) governed by a Cardy-limit approximation, not an exact identity. In every reading, the K3 geometry that appears is an *output* of the charges, via the attractor mechanism, never an input a physical system could dial to select a cosmological parameter – the boundary this project has stated explicitly for other researchers working on K3-based vacuum selection to check before publishing a stronger claim.

What the cosmological extension does not claim. No cosmological number is derived anywhere in this programme: no boson mass, no Ω , no $w(z)$, no lensing amplitude, no string tension. That dark matter is a scalar condensate, that Λ sits on a flux lattice, or that dark energy is a discrete cascade are the hypotheses of the cited literature, not claims of this paper; what is shown is what follows, and what is testable, if they hold. No mapping exists, or is proposed, from this programme’s own toy superfluid simulations to a real cosmological transition rate or Hubble scale – building one by analogy, rather than by a real physical dependency, is exactly the failure mode this programme’s own discipline (check dependencies, not analogy) exists to prevent.

6 Reproducibility and the standard of evidence

Every mathematical claim in §3 and Table 1 that admits a checkable skeleton is stated as a Lean 4 theorem against a pinned Mathlib (tag v4.34.0-rc2) and re-verified by a second, independent kernel (nanoda) via Comparator, so that acceptance does not rest on trusting one implementation of type theory; every module lists, and negative controls exercise, its exact axiom footprint (never more than propositional extensionality, choice, and quotient soundness). As of this writing the library comprises 257 theorems across 30 modules. Two disciplines run throughout: a

literature gate, meaning every citation above was verified against Crossref or by reading the cited source directly before being written down, and an explicit account, in every case, of what the corresponding Lean statement does *not* prove – the $\mathrm{su}(2)_1$ modular S-matrix module does not prove it is the Kac–Peterson matrix of affine Lie algebra representation theory; the level-rank duality module does not prove S,T-matrix or Verlinde-formula preservation; the Montonen–Olive case needed no new Lean file at all, since Mathlib’s own modular-group library already proves the needed stabilizer facts. Where a candidate case was investigated and found not to fit – topological superconductors, three-dimensional Ising duality, AdS/CFT, and the still-open Calabi–Yau self-mirror question – that finding is reported with the same weight as a positive result, following the same convention this programme has used since its first retraction: a failure or an exclusion, honestly reported, is not a defect in the record.

7 Conclusion

A self-dual point is exactly what it says: a fixed point of an exact, checkable, mathematical symmetry. Whether it is also a cause of anything depends entirely on what it is a fixed point *of* – a discrete set of inequivalent, gapped sectors, or a continuous family of equivalent ones – and that question has, in every one of the twelve cases examined here that turn on a genuine self-map, a definite and checkable answer, or an honestly stated absence of one. The duality is never in question; the sector always is. An interventional experiment in a real physical system supports the presupposition the whole criterion needs: that a sector, once identified, is not merely a label but a lever. And where the criterion reaches all the way to cosmology, it makes contact with real, present-day data, not only with toy models – an independent, exact reproduction of a published CMB bound, agreeing with it to within a factor of a few, and a forecast showing that no future CMB temperature mission would meaningfully improve it, both computed on a machine anyone can rent for an afternoon.

What remains open is stated as plainly as what is closed: whether a physically distinguished self-mirror Calabi–Yau threefold exists anywhere in the string landscape; whether a genuinely new full confrontation of the discrete dark-energy cascade against present-day CMB *likelihoods* (not the three-bin toy statistic reproduced here), or against the channels the original authors themselves point to – CMB spectral distortions, pulsar timing arrays – rather than more CMB-temperature data, would sharpen or weaken the discriminating test in §5 (a multi-week undertaking for the likelihood route, by our own estimate, not a multi-day one, regardless of computing hardware); and whether further cross-domain cases – of fifteen candidates investigated so far, three were found not to be instances of the criterion at all, and eleven received a definite verdict – exist that fit the criterion as cleanly as Ising or as informatively as Calabi–Yau’s open question. None of these is claimed here. What is claimed is narrower and, we think, more durable: a criterion, stated once, checked eleven times, and not falsified yet.

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Case		Verdict	Why
Ising (Kramers–Wannier)		Forced	Discrete fixed point; order/disorder sectors inequivalent and gapped
$SO(3)$ Yang–Mills, $\theta = \pi$		Forced	Self-dual point separates a $\mathbb{Z}_2$ TQFT from the trivial phase; anomaly-forced [5]
String-gas cosmology		Preferred equilibrium	Both winding and momentum sectors populated and compete [14]
Compact boson / BKT		Not forced	Continuous line of fixed points; $R = \sqrt{2}$ (self-dual) $\neq 2\sqrt{2}$ (vortex threshold)
Superconductor–insulator		Constraint	Duality pins a resistance <i>if</i> self-dual; vortex/charge proliferation causes the transition [6]
Quantum $\Gamma_0(2)$ /Fricke	Hall,	Organisation	Organises the plateau diagram; Laughlin’s argument causes quantisation [8, 10]
RCFT $(\widehat{su}(2)_1)$ / Verlinde		Not forced	Self-dual radius on a continuous line of RCFT fixed points [15]
Particle-vortex / level-rank (FQHE, $\nu = \frac{1}{2}$)		Constraint	Forces an exact Berry phase and Hall conductance; not itself a transition [7]
$K3 \times T^2$ dyon lattice; D1-D5-P-KK sequel		Organisation	Class number counts sectors at fixed entropy; attractor mechanism causes moduli [9, 16]
Seiberg duality ($\mathcal{N} = 1$)	(4D	Not forced	Conformal window is one continuous phase; $N_c = N_f/2$ not distinguished [17]
Montonen–Olive duality ($\mathcal{N} = 4$ SYM)	S -	Not forced	$\tau = i, \rho$ enhanced-symmetry points on a continuous coupling [18, 19]
Calabi–Yau mirror symmetry		Open question	No citable self-mirror threefold exists; duality itself verified as T-duality [11]
Topological superconductors		Not a duality	K-theory classification, not an exchange [20]
3D Ising / Wegner duality		No self-map	Maps to a <i>different</i> theory (a $\mathbb{Z}_2$ gauge theory) [21]
AdS/CFT		No self-map	Coupling-space equivalence; no conjectured involution [22]

Table 1: Every case examined under the criterion, and why. The last three rows are not verdicts of the criterion; they are cases where the criterion’s question does not apply, distinguished explicitly from a negative answer.